

Summary

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Suppose an object is traveling along a straight line, with displacement given by a function $s(t)$.

- The *average velocity*, v_{av} , of the object between time $t = a$ and time $t = b$ is given by

$$\begin{aligned} v_{av} &= \frac{\Delta s}{\Delta t} \\ &= \frac{s(b) - s(a)}{b - a} \end{aligned}$$

- The *instantaneous velocity* of the object at time $t = a$ is given by

$$\begin{aligned} v(a) &= \lim_{\Delta t \rightarrow 0} \frac{\Delta s}{\Delta t} \\ &= \lim_{t \rightarrow a} \frac{s(t) - s(a)}{t - a} \\ &= \lim_{t \rightarrow a} v_{av}(t) \end{aligned}$$

where $v_{av}(t)$ denotes the average velocity on the interval $[a, t]$ if $t > a$, and $[t, a]$ if $t < a$.